

**BACHELOR OF COMPUTER SCIENCE
FACULTY/SCHOOL OF SCHOOL OF COMPUTER SCIENCE
BINA NUSANTARA UNIVERSITY
JAKARTA**

ASSESSMENT FORM

Course: COMP6048001 - Data Structure

Method of Assessment: Case Study

Semester/Academic Year : 2/2025-2026

Name of Lecturer :

Date :

Class :

Topic : Tries

Group Members :	1
	2
	3
	4
	5
	6
	7
	8

Student Outcomes:

SO 2 - Mampu merancang, mengimplementasikan, dan mengevaluasi solusi berbasis komputasi untuk memenuhi serangkaian persyaratan komputasi dalam konteks ilmu computer

Able to design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of computer science

Learning Objectives:

LObj 2.2 - Mampu mengimplementasikan solusi berbasis komputasi untuk memenuhi serangkaian persyaratan komputasi tertentu dalam konteks ilmu computer

Able to implement a computing-based solution to meet a given set of computing requirements in the context of computer science

Learning Outcomes:

LO 3 - Apply data structures using C

No	Related LO-LOBJ-SO	Assessment criteria	Weight	Excellent (85 - 100)	Good (75-84)	Average (65-74)	Poor (0 - 64)	Score	(Score x Weight)
1	LO 3 – LObj 2.2 – SO 2	Ability to design an algorithm for the problem	80 %	Able to solve 76 - 100% the problem with fully functional feature.	Able to solve 51-75% the problem and lack some features.	Able to solve 26-50% of the problem and lack some features.	Able to solve less than 25% of the problem.		
2	LO 3 – LObj 2.2 – SO 2	Ability to explain the algorithm and make a documentation	20 %	Able to explain the algorithm well. The documentation is good and complete.	Able to give a good explanation of the algorithm. The documentation is good enough.	Able to give an enough explanation of the algorithm The documentation is good enough, but it is not complete.	The explanation the algorithm is bad. The documentation is not complete.		
		Total Score: $\sum(\text{Score} \times \text{Weight})$							

Remarks:

ASSESSMENT METHOD

Instructions:

- This case study is individual project.
 - The case study scoring consist of two parts:
 1. [LO3-LObj 2.2-SO 2, 80 points] Design an algorithm in **cpp** to solve the problem.
 2. [LO3-LObj 2.2-SO 2, 20 points] Create a documentation for your program in PDF (Explain the logic, purpose, and maybe limitation behind each line of the code/function with human language), **please write the code, don't just screenshot it**. Test your program with **custom cases that you made** and put the custom cases along with the result (**screenshots**) in the documentation PDF.
- Example code explanation *may* look like this (**Explaining each line of code**):

```
include<stdio.h> // included for standard input output function

int main(){
    // declare variable number as integer
    int number;
    // Ask for input number from 1 to 10. This code will loop while the number inputted is either less than 1 or more than 10.
    do{
        printf("Enter number between 1 & 10: ");
        scanf("%d", &number);
    }while(number < 1 || number > 10);
    return 0;
}
```

- Example code explanation for function *may also* look like this (**Explaining how a function works**):

```
int biggerNumber(int a, int b){
    int bigger = (a > b) ? a : b;
    return bigger;
}
```

The code above is used to find a bigger number from 2 numbers which are in the parameter: 'a' and 'b'. To find the bigger number, I simply use a ternary operator which has the condition if 'a' is more than 'b', then 'a' is the bigger number, else then 'b' is the bigger

number, I assign either value in a temporary variable 'bigger', then return it. By using this logic, the function will return 'b' if the numbers have the same value, which is okay in our program.

- The custom case (**that you made yourself**) should **at least** have:
 - **Input of 15 slang words,**
 - **Search 5 words,**
 - **View prefix 5 words and**
 - **View all.**
- Submit the **.cpp** and **PDF** files to Binusmaya (zip them).

Note for Lecturers:

- The lecture notifies this case study to the student from Week 1.
- Deadline for the case study *ideally* is in week 9. However, the deadline set in Binusmaya will be in the last week.
- The student should submit the report to binusmaya no later than deadline.
- **If the students do plagiarism, their score for this case study will be 0 (zero). The lecturers have the privilege to determine whether the students do plagiarism.**

Soal

Case

Boogle

Boogle is a company that create, and document new slang words based on the internet. You as a programmer working at the Boogle company are asked to create an application that is useful for seeing what slang words have been released by Boogle. **Ensure that you didn't use regex in your work, or it will affect your score. Please focus on the main logic and main feature! (Design are not scored).** The requirements are:

- The application consists of **5 menus**:
 1. **Release a new slang word**
 2. **Search a slang word**
 3. **View all slang words starting with a certain prefix word**
 4. **View all slang words**
 5. **Exit**
- If user choose menu **1** (“**Release a new slang word**”), then the program will:
 - Ask the user to **input** the **new slang word**. Validate that the **slang word** must be **more than 1 character and contains no space**.
 - Ask the user to **input** the **description (meaning)** of the new word. Validate that the **description** must be **more than 1 word**.
 - **Store** the new released slang word to a **Trie** data structure along with its description.

```
Input a new slang word [Must be more than 1 characters and contains no space]: d
Input a new slang word [Must be more than 1 characters and contains no space]: d a
Input a new slang word [Must be more than 1 characters and contains no space]: da
Input a new slang word description [Must be more than 2 words]: The
Input a new slang word description [Must be more than 2 words]: The word the

Successfully released new slang word.
Press enter to continue...
```

- If the slang word **already exists** in the Trie, then **update** the description with the new description.

```
Input a new slang word [Must be more than 1 characters and contains no space]: da
Input a new slang word description [Must be more than 2 words]: Same meaning as word "the"

Successfully updated a slang word.
Press enter to continue...
```

- If user choose menu **2** ("**Search a slang word**"), then the program will:
 - Ask the user to **input the slang word that want to be searched**. Validate that the **slang word** must be **more than 1 character and contains no space**.
 - **Search** the input word in the Trie data structure.
 - If there **is no such word**, please **show empty message** for the user and go back to main menu.

```
Input a slang word to be searched [Must be more than 1 characters and contains no space]: de

There is no word "de" in the dictionary.
Press enter to continue...
```

- If there **is such word**, please **show the word along with its description**.

```
Input a slang word to be searched [Must be more than 1 characters and contains no space]: d
Input a slang word to be searched [Must be more than 1 characters and contains no space]: d a
Input a slang word to be searched [Must be more than 1 characters and contains no space]: da
```

```
Slang word : da
Description : Same meaning as word "the"
```

```
Press enter to continue...
```

- If user choose menu **3** ("**View all slang words starting with a certain prefix word**"), then the program will:
 - Ask the user to **input the prefix word that want to be searched**.
 - **Search** the input word in the Trie data structure.
 - If there **is no such word**, please **show empty message** for the user.

```
Input a prefix to be searched: de
```

```
There is no prefix "de" in the dictionary.
Press enter to continue...
```

- If there **is such word**, please **show the list of words in the dictionary that starts with the prefix word in lexicographical order**.

Input a prefix to be searched: da

Words starts with “da”:

1. da
2. dadan
3. daijobu
4. dawg
5. dazz

Press enter to continue...

- If user choose menu **4** (“**View all slang words**”), then the program will:
 - If there **is no word yet** in the dictionary, please **show empty message** for the user.

There is no slang word yet in the dictionary.
Press enter to continue...

- Else, please **show the list of all words in the dictionary in lexicographical order**.

List of all slang words in the dictionary:

1. da
2. dadan
3. daijobu
4. dawg
5. dazz
6. rizz
7. simp

Press enter to continue...

- If user choose menu **5** (“**Exit**”), then the program will be closed.

Thank you... Have a nice day :)